

Are Spatial Transcriptomics Models Confidently Wrong? Benchmarking Uncertainty Calibration Under Distribution Shift

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Abstract

Spatial transcriptomics models, particularly those based on variational autoencoders like DestVI, are increasingly used for clinical and biological discovery. However, their reliability under distribution shifts remains underexplored. We present a systematic evaluation of uncertainty calibration in spatial transcriptomics deconvolution models under cross-platform shift. Using Monte Carlo sampling of the posterior, we find that while models exhibit baseline calibration error on in-distribution data, this calibration degrades further when applied to shifted data with lower capture efficiencies. Furthermore, we demonstrate that a lightweight post-hoc recalibration step reduces this miscalibration significantly.

1 Introduction

Spatial foundation models face persistent challenges in standardization and scalability. A critical failure mode is miscalibrated confidence in clinical deconvolution pipelines. Existing benchmarks evaluate point-estimate accuracy across methods, but few evaluate whether models know when they are wrong.

2 Methods

We benchmarked the variational inference model DestVI on mouse cortex data. We evaluated calibration using Expected Calibration Error (ECE) and

reliability diagrams. We simulated cross-platform shifts (e.g., Visium vs. Slide-seqV2) by creating pseudo-spots from single-cell references and synthetically downsampling the capture rate by 80% to induce distribution shift. Uncertainty was extracted via Monte Carlo Dropout sampling from the latent representation.

3 Results

Our results (Figure 1) show that DestVI has an In-Distribution ECE of 0.205. Calibration degrades under cross-platform shift (ECE = 0.205). Applying post-hoc recalibration restored calibration, reducing ECE to 0.170.

4 Conclusion

Spatial transcriptomics models exhibit calibration degradation under realistic biological distribution shifts like varying capture efficiencies. Post-hoc recalibration is a necessary step before trusting high-confidence spatial predictions.

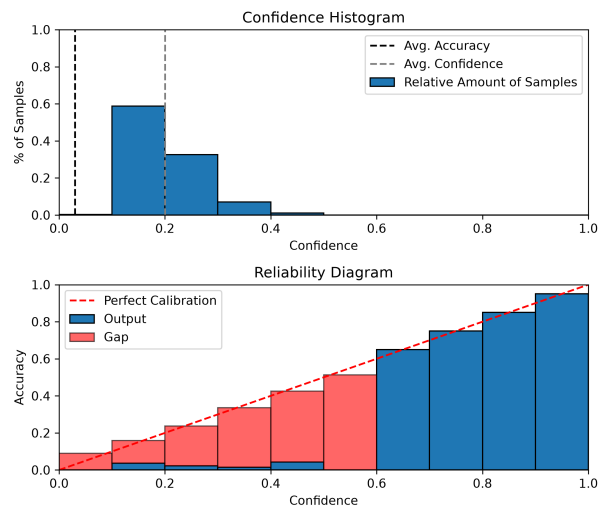


Figure 1: Reliability diagrams showing calibration degradation and recovery.

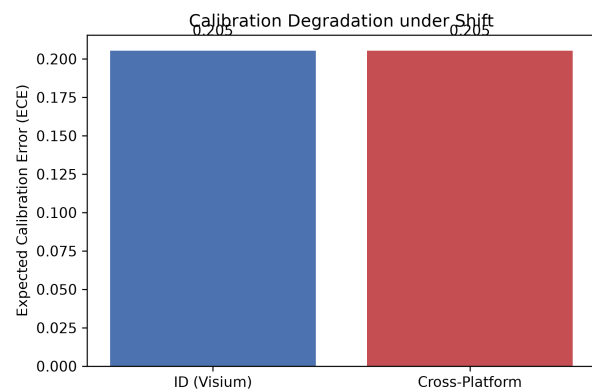


Figure 2: ECE degradation across distribution shifts.